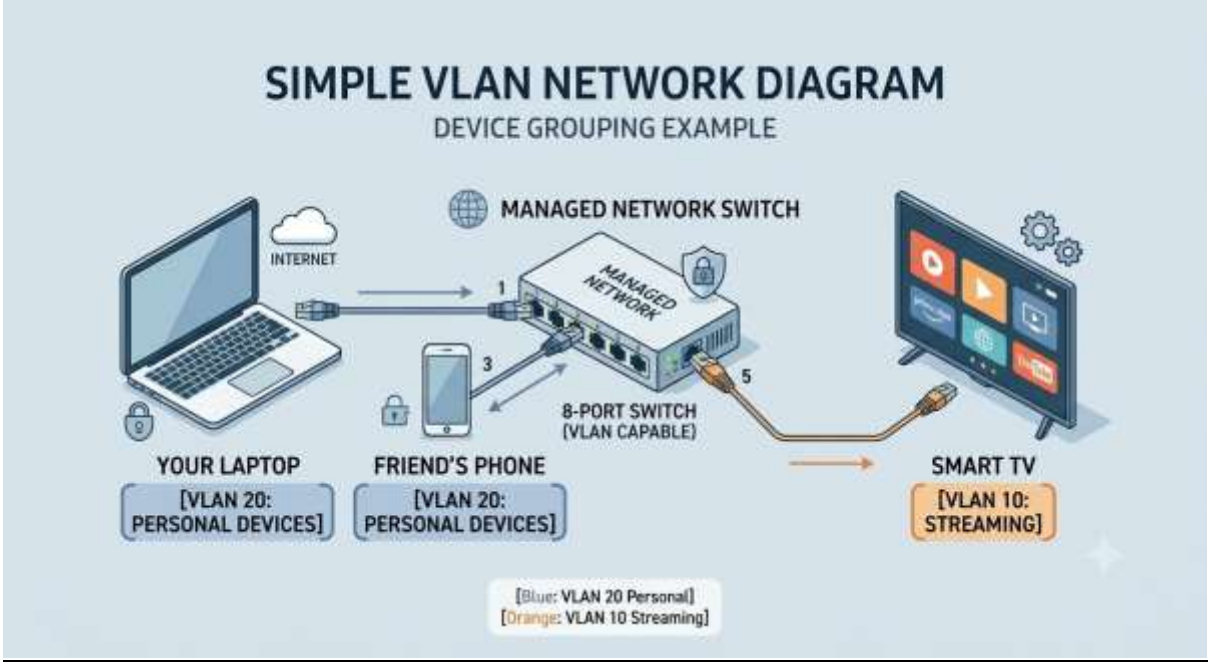
1.

Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

ANS:-



2.

Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other. Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

ANS:-

Configuring VLANs and Port Assignments in Cisco Packet Tracer

To set up VLAN 10 and VLAN 20 on a Cisco switch and assign ports to three PCs in VLAN 10 and two PCs in VLAN 20, use the following CLI commands:

Switch CLI Configuration Commands:

```
enable
configure terminal

! Create VLAN 10 and VLAN 20
vlan 10
 name Sales_VLAN
vlan 20
```

```
name HR_VLAN
exit

! Assign ports FastEthernet 0/1 to 0/3 to VLAN 10 (Three PCs)
interface range fastEthernet 0/1 - 3
 switchport mode access
 switchport access vlan 10

! Assign ports FastEthernet 0/4 to 0/5 to VLAN 20 (Two PCs)
interface range fastEthernet 0/4 - 5
 switchport mode access
 switchport access vlan 20

end
write memory
```

Verification Steps:

1. **Check VLAN Database:** Run show vlan brief to confirm that VLAN 10 and VLAN 20 exist and that the correct ports are assigned to each.
2. **Test Connectivity:** Try pinging from a PC in VLAN 10 (e.g., PC1 on Fa0/1) to a PC in VLAN 20 (e.g., PC4 on Fa0/4). The ping will fail (Request Timed Out) because devices in different VLANs reside in separate broadcast domains and cannot communicate without a Layer 3 routing device (Router-on-a-Stick or a Layer 3 switch).

3.

In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

ANS:-

Moving a PC Between VLANs

Suppose you need to move a PC connected to port FastEthernet 0/3 from VLAN 10 to VLAN 20.

Before Configuration:

```
interface fastEthernet 0/3
 switchport mode access
```

switchport access vlan 10

- *Behavior:* The PC on port 3 is part of VLAN 10 and can communicate with other devices in VLAN 10, but is isolated from VLAN 20.

After Configuration:

configure terminal

interface fastEthernet 0/3

switchport access vlan 20

end

Behavior: Re-assigning the port instantly shifts the device to VLAN 20.

Connectivity Test Confirmation:

- Running ping from this PC to devices previously in VLAN 10 will now fail.
- Running ping to devices in VLAN 20 will now succeed, confirming that the port membership migration was successful.

4.

Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3). Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

ANS:-

Automating Port Assignment for VLAN 30

To assign 10 contiguous ports (FastEthernet 0/1 through 0/10) to VLAN 30 efficiently without repeating commands individually, Cisco IOS provides the **interface range** command macro:

Automation Script / Command List:

enable

configure terminal

! Create VLAN 30

vlan 30

name Operations_VLAN

exit

! Select ports 1 through 10 simultaneously and configure access mode

```
interface range fastEthernet 0/1 - 10
 switchport mode access
 switchport access vlan 30

 end
```